

Брунн–Минковский и смешанные объёмы

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A single inequality from which isoperimetry follows in every dimension. On the way mixed volumes appear: the volume of a Minkowski sum is a polynomial, and its coefficients are themselves geometric quantities.

Theorem 1 (Brunn 1887, Minkowski 1896). For nonempty compact $A, B \subseteq \mathbb{R}^n$

$$|A + B|^{1/n} \geq |A|^{1/n} + |B|^{1/n},$$

with equality (for convex bodies of positive volume) exactly when A and B are homothetic.

Theorem 2 (Steiner formula). $|K + tB_2^n| = \sum_{i=0}^n \binom{n}{i} W_i(K) t^i$, where $W_0(K) = |K|$, $nW_1(K)$ is the surface area and $W_n(K) = \omega_n$.

1. (Putnam 2003) Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be nonnegative reals. Prove

$$(a_1 a_2 \cdots a_n)^{1/n} + (b_1 b_2 \cdots b_n)^{1/n} \leq ((a_1 + b_1)(a_2 + b_2) \cdots (a_n + b_n))^{1/n}.$$

Observe that this is exactly the Brunn–Minkowski inequality for boxes with sides a_i and b_i .

2. (Hadwiger–Ohmann 1956) Prove Theorem 1 in full: first for finite unions of boxes (cut by a coordinate hyperplane so that the volume ratio is preserved and induct on the number of boxes), then in general by approximation.
3. Deduce the isoperimetric inequality in $\mathbb{R}^n$: for a convex body K ,

$$\frac{S(K)}{S(B_2^n)} \geq \left(\frac{|K|}{|B_2^n|} \right)^{(n-1)/n}.$$

(Hint: $S(K) = \lim_{t \rightarrow 0^+} (|K + tB| - |K|)/t$.)

4. Prove that $t \mapsto |K + tB_2^n|$ is a polynomial of degree n ; compute its coefficients for a square, a triangle and a disc in $\mathbb{R}^2$ and for a cube in $\mathbb{R}^3$, and say which geometric quantities appear.
5. (a) (VJIMC 2003, cat. II) Let $D_1, \dots, D_n$ be discs in the plane and let $a_{ij} = |D_i \cap D_j|$. Prove that $\sum_{i,j} a_{ij} x_i x_j \geq 0$ for all real $x_1, \dots, x_n$.
 (b) (Капачёв 3.55) Prove that the entrywise product of two positive definite symmetric matrices is positive definite, and explain in what sense (a) is a continuous version of (b).
6. Prove Minkowski's first inequality $V(K, \dots, K, L) \geq |K|^{(n-1)/n} |L|^{1/n}$ from Brunn–Minkowski, and show that it contains the isoperimetric inequality as the case $L = B_2^n$.